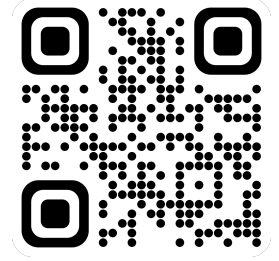


Zeel B Patel

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Email (personal): zeelpatel4896@gmail.com

GitHub: <https://github.com/patel-zeel>
LinkedIn: [zeel-b-patel](#)



EDUCATION

- **PhD in Computer Science,** *Jan 2020 - Feb 2026*
Thesis: AI for Air Quality: Modeling, Sensor Placement and Source Detection
Thesis Defended: 4th Feb 2026
Advisor: [Nipun Batra](#)
CPI: 9.7/10
IIT Gandhinagar, Gujarat, India
- **M.Tech. in Smart Manufacturing,** *Aug 2017 - May 2019*
CGPA: 9.17/10
IIITDM Kancheepuram, Chennai, India
- **B.E. in Mechanical Engineering,** *Aug 2013 - May 2017*
CGPA: 7.52/10.0
Kadi Sarva Vishwavidyalaya, Gandhinagar, India

WORK EXPERIENCE

- **Postdoctoral Researcher, Microsoft Research India** *Jul 2025 - Present*
Project: Latency-bound accurate retrieval
- **Software Engineering Intern, Google Summer of Code** *Jun 2022 - Sep 2022*
Mentor: [Kevin P Murphy](#)
Project: Contribute code and content to code-backed Probabilistic ML textbook
GitHub repo: <https://github.com/probml/pyprobml>
Final report: <https://patel-zeel.github.io/gsoc22>
- **Data Scientist, R&D Team** *Jun 2019 - Dec 2019*
Inspirisys Solutions Ltd., Chennai, India

OPEN SOURCE LIBRARIES

BIJAX (7★): Bayesian Inference in JAX; used in Kevin Murphy's Probabilistic Machine Learning book

GPAX (6★): Gaussian processes in JAX, developed for my PhD research

skgpytorch (3★): Scikit-learn interface for GPyTorch enabling rapid GP model prototyping

ASTRA (3★): AI for Sustainability Toolkit, developed for my research group at IIT Gandhinagar

SELECTED OPEN SOURCE CONTRIBUTIONS

GPyTorch: Added probabilistic metrics module; essential for proper evaluation of probabilistic models beyond RMSE ([PR #1870](#))

supervision (Roboflow): Added Oriented Bounding Boxes support to this popular object detection library ([PR #1502](#))

matplotlib: Fixed slow plotting for PyTorch/JAX arrays; required cross-library coordination (NumPy, Matplotlib, PyTorch, JAX) ([PR #25887](#))

Stheno: Implemented FITC, one of the most widely-used sparse Gaussian process methods ([PR #17](#))

PUBLICATIONS (GOOGLE SCHOLAR PROFILE)

Journal articles

1. **Zeel B Patel**, Rishabh Mondal, Shataxi Dubey, Suraj Jaiswal, Sarath Guttikunda, Nipun Batra
Space to Policy: Scalable Brick Kiln Detection and Automatic Compliance Monitoring with Geospatial Data
Published in ACM Journal on Computing and Sustainable Societies and presented at ACM COMPASS 2025 conference.
Project page: <https://sustainability-lab.github.io/brick-kilns/>

Selected peer-reviewed articles

1. **Zeel B Patel**, Vinayak Rana, Nipun Batra
Scalable Air-Quality Sensor Placement via Gradient-Based Mutual Information Maximization
AAAI 2026 (CORE A*, Oral Presentation - 24.1% acceptance rate)
GitHub repo: <https://github.com/sustainability-lab/gdmi-aqs>
2. **Zeel B Patel**, Palak Purohit, Harsh M Patel, Shivam Sahni, Nipun Batra
Accurate and Scalable Gaussian Processes for Fine-grained Air Quality Inference
AAAI 2022 (CORE A* - 15% acceptance rate)
GitHub repo: <https://github.com/patel-zeel/AAAI22>
3. Sachin Chauhan, **Zeel B Patel**, Sayan Ranu, Rijurekha Sen, Nipun Batra
AirDelhi: Fine-Grained Spatio-Temporal Particulate Matter Dataset From Delhi For ML based Modeling
In NeurIPS 2023 Datasets and Benchmarks (CORE A* - 32.7% acceptance rate)
4. Rishabh Mondal, Jeet Parab, Heer Kubadia, Shataxi Dubey, Shardul Junagade, **Zeel B Patel**, Nipun Batra
SentinelKilnDB: A Large-Scale Dataset and Benchmark for OBB Brick Kiln Detection in South Asia Using Satellite Imagery
NeurIPS 2025 Datasets and Benchmarks (CORE A*)
Project page: <https://sustainability-lab.github.io/sentinelkilndb/>
5. Rishiraj Adhikary, **Zeel B Patel**, Tanmay Srivastava, Nipun Batra, Mayank Singh, Udit Bhatia, Sarath Guttikunda
Vartalaap: What Drives #AirQuality Discussions: Politics, Pollution or Pseudo-science?
CSCW 2021 (CORE A)
GitHub repo: <https://github.com/rishi-a/Vartalaap>
6. Karm Patel, Rishiraj Adhikary, **Zeel B Patel**, Nipun Batra, Sarath Guttikunda
Samachar: Print News Media on Air Pollution in India
COMPASS 2022
GitHub repo: <https://github.com/karm-patel/Samachar-News-media-on-air-pollution>

BOOK CONTRIBUTIONS

- **Probabilistic Machine Learning: Advanced Topics:** <https://probml.github.io/pml-book/book2.html>
Co-authored Section 34.7 (Active Learning) with Kevin Murphy

AWARDS

Awards

- Microsoft Research India PhD Award. Unrestricted grant of 1 million INR. 2024
- Outstanding Graduate Teaching Fellow award in Probabilistic Machine Learning course, IIT Gandhinagar. Fall 2022

Contributed to grant applications

- Google PaliGemma Compute Grant 2024 (\$5000 credits in Google Cloud Platform)
- Google Compute Grant 2021 (\$5000 credits in Google Cloud Platform)

TEACHING EXPERIENCE

Graduate Teaching Fellow (teaching a course along with the instructor)

- **Probabilistic Machine Learning**
IIT Gandhinagar Fall 2022

Teaching Assistant

- **Probabilistic Machine Learning**
IIT Gandhinagar *Fall 2023*
- **Machine Learning**
IIT Gandhinagar *Spring 2023*
- **Machine Learning**
IIT Gandhinagar *Spring 2022*

Guest lectures

- **Introduction to Active Learning**
Ubiquitous computing, IIT Gandhinagar *Fall 2021*
- **Introduction to Bayesian Machine Learning**
Machine Learning, IIT Gandhinagar *Spring 2021*

INVITED TALKS

Google Global Air Quality Summit

14th Nov, 2024

Topic: AirChat: LLM-Powered Chatbot System to Democratize Urban Air Quality Information

App Demo: <https://sustainabilitylabiitgn-vayubuddy.hf.space>

Gurugram, India

Air Sensors International Conference

26th Aug, 2022

Topic: Accurate and Scalable Gaussian Processes for Fine-grained Air Quality Inference

Organized by [CSTEP, India](#) and [UC Davis](#)

Bengaluru, India